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Integration Rules Cheat Sheet

Complete integration rules cheat sheet with all antiderivative formulas, substitution, integration by parts, and trigonometric integrals. Free PDF download.

Basic Integration Rules

NAME	FORMULA	NOTES
Constant	$\int k\,dx = kx + C$	
Power Rule	$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$	Valid for n ≠ -1. Add 1 to exponent, divide by new exponent
Sum Rule	$\int (f(x) + g(x))\,dx = \int f(x)\,dx + \int g(x)\,dx$	
Constant Multiple	$\int k f(x)\,dx = k \int f(x)\,dx$	

Common Integrals

NAME	FORMULA	NOTES
1/x	$\int \frac{1}{x}\,dx = \ln\lvert x\rvert + C$	
Exponential	$\int e^x\,dx = e^x + C$	
General Exponential	$\int a^x\,dx = \frac{a^x}{\ln(a)} + C$	

Trigonometric Integrals

NAME	FORMULA	NOTES
Sine	$\int \sin(x)\,dx = -\cos(x) + C$	
Cosine	$\int \cos(x)\,dx = \sin(x) + C$	
Secant Squared	$\int \sec^2(x)\,dx = \tan(x) + C$	
Cosecant Squared	$\int \csc^2(x)\,dx = -\cot(x) + C$	
Secant Tangent	$\int \sec(x)\tan(x)\,dx = \sec(x) + C$	
Cosecant Cotangent	$\int \csc(x)\cot(x)\,dx = -\csc(x) + C$	
Tangent	$\int \tan(x)\,dx = -\ln\lvert\cos(x)\rvert + C$	
Secant	$\int \sec(x)\,dx = \ln\lvert\sec(x) + \tan(x)\rvert + C$	

Inverse Trig Integrals

NAME	FORMULA	NOTES
Arcsine Form	$\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \arcsin\left(\frac{x}{a}\right) + C$	
Arctangent Form	$\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$	

Integration Techniques

NAME	FORMULA	NOTES
U-Substitution	$\int f(g(x))g'(x)\,dx = \int f(u)\,du$	Let u = g(x), then du = g'(x)dx
Integration by Parts	$\int u\,dv = uv - \int v\,du$	Choose u and dv wisely (LIATE rule)

Definite Integrals

NAME	FORMULA	NOTES
Fundamental Theorem	$\int_a^b f(x)\,dx = F(b) - F(a)$	Where F'(x) = f(x)
Area Between Curves	$A = \int_a^b (f(x) - g(x))\,dx$	Top minus bottom